

Vineeth Avti

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GenAI Engineer

PROFESSIONAL SUMMARY:

- GenAI Engineer with 5+ years of experience building enterprise AI platforms, intelligent automation systems, and scalable machine learning solutions across healthcare, banking, and retail domains.
- Strong analytical and quantitative problem-solving skills analyzing user behavior, retention, engagement, and conversion metrics to uncover growth opportunities and actionable insights.
- Experienced in building dashboards and recurring reports using SQL and spreadsheet tools to track product launches, experiments, and marketing campaign effectiveness.
- Skilled in translating unclear product questions into structured data requirements, developing measurement plans, and defining meaningful business and growth metrics.
- Proficient in Python for data analysis, visualization, and turning ambiguous information into clear, communicable insights for both technical and non-technical stakeholders.
- Strong attention to detail and ability to identify and question anomalies or unusual trends in large datasets to support data-driven decision making.
- Experienced collaborating across Product, Growth, Marketing, and Engineering teams to establish reliable data practices and scalable analytics workflows.
- Capable of working in fast-moving startup environments, quickly adapting to emerging data technologies to improve customer behavior analysis and operational efficiency.
- Good communication and presentation skills, capable of clearly explaining analytical findings and recommendations to diverse teams for impactful decision support.
- Interest and experience analyzing e-commerce, marketplaces, and consumer product data to optimize user onboarding, purchases, and retention strategies.

TECHNICAL SKILLS:

- **Generative AI & LLM Engineering** - Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Prompt Engineering, Semantic Search, Conversational AI, AI Agents, Contextual Retrieval, Prompt Optimization, Knowledge Retrieval Systems, Reranking
- **GenAI Frameworks & Vector Technologies** - LangChain, LangGraph, LlamaIndex, Azure OpenAI, AWS Bedrock, OpenAI APIs, Hugging Face Transformers, FAISS, Pinecone, ChromaDB, Cohere Rerank, Vector Databases
- **Machine Learning & NLP** - Machine Learning Pipelines, NLP, Text Embeddings, Semantic Similarity, Recommendation Systems, Predictive Analytics, Feature Engineering, Text Classification, Summarization
- **Programming & AI Development** - Python, SQL, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, REST APIs, API Integrations, Data Processing Automation
- **Cloud Platforms & AI Services** - Microsoft Azure (Azure OpenAI, Azure ML, Databricks, ADLS Gen2, Azure Data Factory), AWS (AWS Bedrock, SageMaker, Lambda, S3, CloudWatch)
- **Data Engineering & Distributed Processing** - PySpark, Apache Spark, ETL/ELT, Distributed Data Processing, Batch Processing, Delta Lake, Data Pipelines, Data Integration
- **AI Orchestration & MLOps** - Kubeflow Pipelines, MLflow, Airflow, Workflow Automation, LLMOps, Model Registry, Containerized Inference, Docker, Kubernetes, CI/CD for ML Pipelines
- **Monitoring, Evaluation & Governance** - RAGAS Evaluation, Prompt Evaluation, Prompt Versioning, Hallucination Reduction, Guardrails, Observability, Model Drift Monitoring, Response Validation, Weights & Biases (W&B)
- **Security & Compliance** - RBAC, Secure AI Access, Data Encryption, Responsible AI Practices, Governance Standards, Enterprise Security Controls
- **Others** - Excel, Spreadsheets, Data Visualization Tools, Dashboard Building, Recurring Reporting, Experiment Analysis, Product Launch Analysis, Marketing Campaign Analysis, Metrics Definition, Measurement Plans

PROFESSIONAL EXPERIENCE:

Humana Inc.

January 2024 – Present

GenAI Engineer

Responsibilities:

- Analyze healthcare knowledge retrieval to identify patterns supporting AI-assisted conversational search and document assistance in enterprise platforms.
- Build and maintain dashboards measuring AI interaction quality, retrieval precision, and semantic relevance for healthcare data applications.
- Investigate and optimize embedding search accuracy and response grounding improving inference latency across medical AI-assisted workflows.
- Design scalable data pipelines integrating clinical datasets with AI workflows enabling actionable insights for healthcare knowledge discovery.
- Develop prompt orchestration and retrieval strategies to enhance semantic ranking and improve response quality in healthcare AI systems.
- Collaborate with product and engineering teams to define metrics and measurement plans impacting AI-assisted healthcare operational effectiveness.
- Employ RBAC and audit logging ensuring security and compliance aligning AI governance with healthcare data management standards.
- Monitor anomaly detection and retrieval benchmarking pipelines to identify unusual trends and improve reliability of healthcare AI responses.
- Transform unstructured healthcare documents into semantic embeddings enabling precise search and actionable insights within clinical AI environments.
- Communicate complex AI evaluation metrics clearly to healthcare stakeholders supporting product improvement and AI adoption decisions.

Bank of America**February 2021 – December 2022****AI/ML Engineer****Responsibilities:**

- Analyze transactional and fraud detection datasets to identify behavioral trends supporting banking risk mitigation and AI-assisted decision making.
- Build and maintain financial reporting dashboards visualizing key metrics on customer servicing, transaction monitoring, and fraud analytics.
- Interpret semantic analysis and summarization results improving contextual inference workflows supporting secure banking application productivity.
- Investigate features and data anomalies improving model accuracy and identifying unusual user behavior across enterprise financial ML pipelines.
- Collaborate with product and growth teams defining measurement plans for AI services impacting fraud detection and customer servicing improvements.
- Develop MLOps pipelines supporting automated deployment, monitoring, drift detection, and scheduled inference reporting across banking AI environments.
- Transform raw financial datasets using Python and Spark enabling reliable data readiness and enhanced machine learning model performance.
- Communicate evaluation results and data insights clearly to non-technical stakeholders improving AI-assisted enterprise banking workflows.
- Implement encryption and audit controls maintaining secure access and compliance adherence across financial machine learning systems.
- Optimize inference and feature selection workflows reducing latency and improving model predictions in enterprise fraud and banking analytics.

Target Corporation**August 2019 – January 2021****Machine Learning Engineer****Responsibilities:**

- Analyze customer behavior and clickstream datasets to identify product personalization opportunities and optimize retail recommendation effectiveness.
- Build dashboards tracking user engagement, retention, and sales metrics supporting e-commerce product and marketing decision frameworks.
- Investigate data quality issues and transformation workflows ensuring reliable retail analytics supporting recommendation workflows and customer insights.

- Implement scalable Spark processing pipelines enabling analysis of large transactional datasets to inform targeted marketing and personalization.
- Collaborate with product teams to define KPIs and measurement strategies improving understanding of user drop-offs and conversion drivers.
- Develop monitoring and validation systems to detect unusual trends or anomalies in customer purchase and engagement datasets.
- Utilize Python and SQL to preprocess and engineer features improving machine learning readiness and enhancing recommendation system accuracy.
- Communicate analytical findings to non-technical teams enabling actionable insights and informed decisions for retail product optimization.
- Support integration of multiple retail data sources into unified analytics platforms enhancing visibility over user and sales behaviors.
- Conduct evaluation of recommendation models and experiment results to measure effectiveness of marketing campaigns and product launches.

Educational Details:

Master of Science in Applied Security & Analytics - University of Findlay

Bachelor of Technology in Electronics & Communication Engineering - Jawaharlal Nehru Technological University Hyderabad